LLM exam sprint

Prepared for 28 October 2026 • Based on the uploaded lecture set • Start: 25 October

Your goal: explain the core ideas, distinguish similar methods, and solve small calculations without looking at the slides. This pack is a first-pass revision aid, not evidence of complete official syllabus coverage.

A provisional 10-hour LLM budget

Use 12 blocks of 50 minutes, with a 10-minute break between blocks. Spread them across 25-27 October. Reserve the rest of your study time for the other two exams; their subjects and timings are still needed to finalize the shared plan.

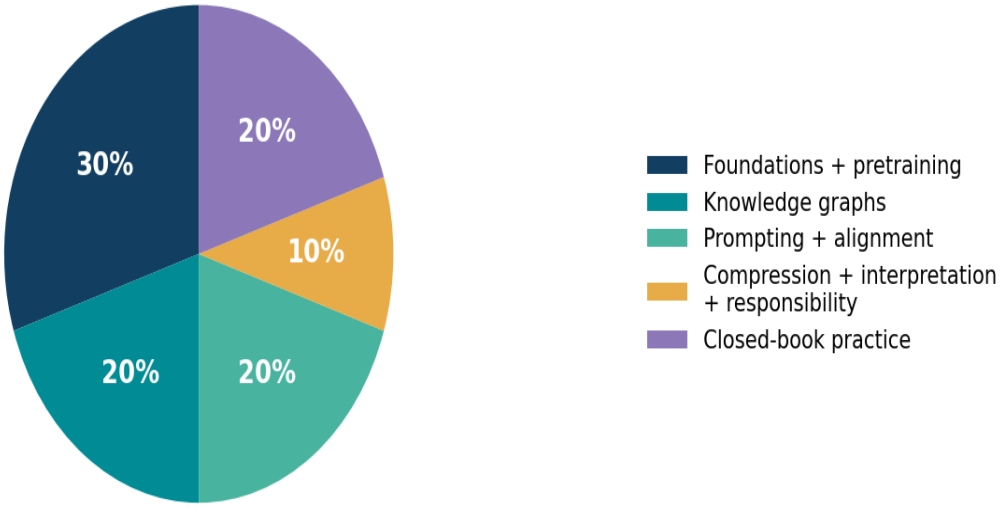


Chart meaning: suggested study minutes, not predicted exam marks. Priorities reflect prerequisites and opportunities for recall and numerical practice. Adjust after seeing your assignment questions.

Date	LLM blocks	What to finish
25 Oct	4 × 50 min	Tokenization; seq2seq/decoding; attention; Transformer positions and code.
26 Oct	4 × 50 min	Pretraining/prompting; alignment; KG completion/models; temporal KGs/hierarchies.
27 Oct	4 × 50 min	Compression/interpretability/responsibility; numericals; practice; repair errors.
28 Oct	20-30 min if feasible	Use the final-page checklist. Match review time to your actual exam session.

How to use the pack

The 50-minute loop

25 minutes: learn one topic. 10 minutes: close the notes and explain it aloud. 10 minutes: solve one calculation or comparison. 5 minutes: record mistakes and the corrected rule. Revisit the same mistakes the next day.

If only five hours remain

Use 90 minutes for tokenization, decoding and attention; 45 for Transformer architecture and pretraining; 45 for alignment; 45 for KG completion, models, hierarchies and time; 30 for compression, interpretability and responsibility; 45 for calculations and error repair. This is a fallback with shallower coverage, not equivalent mastery.

Lecture map and where to focus

Uploaded file	Core content	Pack page
Lec_9.pdf	BPE, WordPiece, Unigram	3
Lec_12.pdf; Lec_13.pdf	Seq2seq; greedy/beam/top-k/top-p/temperature	4
Lec_14.pdf	Attention and bottleneck	5
Lec_16.pdf	Positions, RoPE, residuals, normalization	6
Lec_17.ipynb; Lec_19.pdf	Transformer shapes and pretraining	7
Lec_23.pdf	Advanced prompting and sensitivity	8
Lec_24.pdf; Lec_25.pdf	Reward models, REINFORCE, PPO	9-10
Lec_27.pdf; Lec_28.pdf	KG completion/evaluation; translation/rotation	11-12
04_Knowledge_6_Temporal.pptx.pdf; Lec_34.pdf	Temporal KGs; hierarchies	13
Lec_30.pdf	Quantization, pruning, distillation	14
Lec_31.pdf; Lec_32.pdf	Residual stream and interpretability	15
Lec_36.pdf	Responsible LLMs and bias mitigation	16

Lec_36 - Copy.pdf has identical extracted text to Lec_36.pdf and is counted once. There are 17 distinct PDFs plus one notebook in the working study set. The screenshot identifies the course as Introduction to Large Language Models; it does not establish the exam pattern.

Scope rule: the numbered lectures have gaps. Treat these uploads as the supplied revision scope, not proof that every assessed lecture has been uploaded. If available, use official scope and weekly assignments to refine this plan.

1 Tokenization

Source: Lec_9.pdf, especially pp. 4-14, 16-28, 30-44.

Tokenization splits text into units that receive token IDs. A token may be a word, part of a word, character, or byte. Subword methods balance vocabulary size against sequence length and help represent rare words.

Method	Training idea	Tokenization / key trap
BPE	Repeatedly merge the most frequent adjacent token pair; count corpus frequencies.	Apply learned merges in their learned order. Byte-level BPE starts from 256 bytes.
WordPiece	In the lecture formulation, merge the pair with highest frequency ratio.	Longest-match-first using the final vocabulary. BERT uses ## for continuation pieces.
Unigram LM	Start with a large candidate vocabulary; estimate token probabilities and remove low-value pieces.	Choose a high-probability segmentation; dynamic programming/Viterbi avoids enumerating every split.

WordPiece score(a,b) = freq(a,b) / [freq(a) × freq(b)]

Unigram segmentation score = product of the probabilities of its pieces.

Worked BPE step

Corpus: cat ×10, bat ×5, bag ×12, tag ×4, cats ×5. With character tokens, pair (a,t) occurs 10+5+5 = 20 times; (a,g) occurs 12+4 = 16. Merge a+t first. Recompute pair counts after every merge.

Worked WordPiece decision

Pair A: frequency 20, component counts 100 and 50 → score 0.004. Pair B: frequency 6, component counts 10 and 10 → score 0.06. WordPiece selects B, despite its lower raw pair frequency. This is the lecture scoring rule, not a universal claim about every implementation.

Worked Unigram decision

P(r)=0.02, P(u)=0.2, P(n)=0.17, P(ru)=0.03, P(un)=0.12. For run: [r,u,n]=0.00068; [ru,n]=0.0051; [r,un]=0.0024. The best segmentation is [ru,n].

Traps: BPE grows vocabulary by merging; Unigram shrinks a seed vocabulary. Under the lecture WordPiece procedure, an unsegmentable word becomes [UNK]. Character-level BPE may encounter unknown characters; byte-level BPE represents input through bytes. SentencePiece is a tokenizer framework supporting algorithms such as Unigram and BPE, rather than one unique algorithm.

Close the notes and retrieve

Why use subwords? How does BPE choose a merge? How does WordPiece choose one? What does Unigram maximize for a segmentation?

2 Seq2seq and decoding

Sources: Lec_12.pdf pp. 3-12, 27-28; Lec_13.pdf pp. 2-17.

Seq2seq: an encoder represents an input sequence and a decoder generates the output, conditioned on the input and preceding output tokens. Tasks include translation, summarization and dialogue. A fixed-vector RNN encoder creates an information bottleneck for long inputs.

P(y | x) = product over t of P(y_t | x, y_<t)
Sequence log-score = sum over t of log P(y_t | x, y_<t)

Strategy	Rule	Distinction
Greedy	Choose the highest-probability next token.	Local best choice can miss the best complete sequence.
Beam search	Keep k highest-scoring partial sequences.	Deterministic search under fixed settings; no guarantee of global optimum.
Top-k sampling	Keep k tokens, renormalize, sample.	Fixed candidate count; k=1 gives greedy.
Top-p / nucleus	Keep the smallest sorted set whose cumulative mass is at least p; renormalize, sample.	Candidate count adapts to the distribution.
Temperature	Use softmax(logits / T), T > 0.	Low T sharpens; high T flattens. Greedy argmax is unchanged by positive T.

Two calculations you must be able to do

Probabilities: A=.50, B=.25, C=.15, D=.10. Top-k with k=2 retains A,B and yields probabilities 2/3, 1/3. Top-p with p=.80 retains A,B,C because .75 is insufficient and .90 reaches the threshold. New probabilities: .556, .278, .167.

First-token probabilities A=.60, B=.40. Best continuation after A has probability .50; after B, .90. Greedy follows A and reaches .60×.50=.30. A beam retaining both prefixes can discover B’s sequence with .40×.90=.36.

Length and stopping

Multiplying probabilities usually reduces the score as length grows, so raw sequence likelihood can favor short outputs. Length normalization helps compare completed hypotheses. Ended hypotheses are stored; remaining beams continue until the stopping rule is met. Exhaustive search over length L costs exponentially many candidates, $|V|^L$.

Close the notes and retrieve

Beam search versus top-k: what is kept, and is the next choice sampled? Why does top-p use a variable number of tokens?

3 Attention: understand the calculation

Sources: Lec_14.pdf pp. 5-28; Lec_31.pdf pp. 2-3; Lec_17.ipynb.

Intuition: when producing a token, the model asks which available representations matter now. Attention builds a weighted mixture of value vectors. In seq2seq it lets each decoder step access encoder states directly.

Symbol	Meaning	Where it comes from
Q: query	What information am I looking for?	Current representation / decoder state.
K: key	What can each position match?	Representations being attended to.
V: value	What information will be combined?	Representations being attended to.

$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T / \sqrt{d_k} + M)V$$

$Q = XW_Q$, $K = XW_K$, $V = XW_V$ for self-attention.

Order: project → compare queries with keys → scale → mask if required → softmax across key positions → weighted sum of values. Scaling by $\sqrt{d_k}$ limits large dot products and softmax saturation.

A fully worked attention row

Let q=(1,0), k1=(1,0), k2=(0,1), d_k=2; v1=(2,0), v2=(0,4). Raw scores are (1,0); scaled scores are (0.707,0). Softmax gives approximately (.670,.330). Output = .670(2,0)+.330(0,4) = (1.340,1.321), allowing rounding.

Masking: a frequent conceptual trap

For causal attention, allowed positions $j \leq i$ receive mask value 0; future positions $j > i$ receive $-\infty$. After softmax their weights are 0. Set forbidden logits to $-\infty$, not 0: exp(0)=1 and would leave probability mass on forbidden positions. A padding mask hides padding; a causal mask hides future tokens.

Self-attention versus cross-attention

Self-attention: Q,K,V are projections of the same sequence. Cross-attention: Q comes from the decoder; K,V come from encoder outputs. Decoder masked self-attention may include the current input position, because its prediction target is the following token.

Multi-head: different learned projections attend in parallel; concatenate head outputs and project them. With d_{model}=512 and 8 equal heads, each head has dimension 64. Dense self-attention has an $n \times n$ score matrix per head: doubling length gives four times as many score entries.

Close the notes and retrieve

Write the attention equation, explain every symbol, and calculate one softmax row. Why is the mask applied before softmax?

4 Transformer components and positions

Source: Lec_16.pdf; supporting attention details from Lec_17.ipynb and Lec_31.pdf.

Encoder block: self-attention plus feed-forward network, with residual connections and normalization. Decoder block: causal self-attention; encoder-decoder cross-attention in an encoder-decoder model; feed-forward network. Decoder-only LMs omit encoder cross-attention.

Component	Purpose	Recall point
Position information	Make order available to attention.	Self-attention alone does not supply sequence order.
Residual connection	Add a block update to its input.	x + F(x); assists information and gradient flow.
Layer normalization	Normalize hidden features within a token representation.	Not across the batch; includes learned scale and shift.
Feed-forward network	Transform features at each position.	Same learned position-wise network applied to tokens.

$$PE(pos,2i) = \sin(pos / 10000^{2i/d_model})$$
$$PE(pos,2i+1) = \cos(pos / 10000^{2i/d_model})$$
$$LayerNorm(x) = \gamma \times (x-mean(x))/\sqrt{(variance(x)+\varepsilon)} + \beta$$

Absolute, relative and rotary positions

Absolute: associate each position with an embedding, learned or sinusoidal; standard sinusoidal encodings are added to token embeddings. Relative: attention depends explicitly on position differences, for example through a relative bias. RoPE: rotate paired dimensions of Q and K by position-dependent angles. Their dot product incorporates relative displacement.

RoPE memory hook: shift both positions by the same amount → rotate both by the same extra angle → their relative angle and dot product are preserved. RoPE is multiplicative rotation, rather than simply adding a position vector.

Small checks

At pos=0, each sine component is 0 and each cosine component is 1. For a two-feature vector (1,3), mean=2 and population variance=1; normalized values are approximately (−1,+1), before learned γ and β and the small ε adjustment.

Traps: position encodings being mathematically defined beyond training length does not guarantee good long-context performance. LayerNorm is not min-max scaling. Some models normalize before sublayers and others after; know the arrangement used in the supplied notebook.

Close the notes and retrieve

Sketch encoder versus decoder components. Explain how RoPE represents relative position and why residual additions must preserve compatible dimensions.

5 Pretraining and notebook essentials

Sources: Lec_19.pdf pp. 2-19, 31-42, 49-50; Lec_17.ipynb.

Family	Context / objective	Examples
Encoder-only	Bidirectional representations; masked-token prediction in BERT.	BERT
Decoder-only	Causal context; predict the next token.	GPT, Llama
Encoder-decoder	Encoder sees input; causal decoder generates conditioned output.	BART, T5

BART: corrupt text by masking, deleting or reordering parts, then reconstruct the full clean sequence. T5: text-to-text formulation; in span-corruption pretraining, replace spans with distinct sentinel tokens and predict the removed spans with sentinels. C4 expands to Colossal Clean Crawled Corpus.

Next-token loss = −sum over t of log Pθ(x_t | x_<t)
For a target given probability .25, its negative log-loss is −ln(.25) ≈ 1.386.

Training versus generation: teacher forcing uses the true preceding target tokens during training. During autoregressive generation, the model uses its own preceding outputs. The causal mask allows next-token training at many positions in parallel while preventing future-token leakage.

Read the uploaded notebook by shapes

Notebook operation	Shape / meaning
Input IDs	(N, length): integer IDs for nn.Embedding. The initial floating tensors are illustrative; the final inference example uses integer tensors.
Split Q,K,V into heads	(N, length, heads, head_dim), with embed_size divisible by heads.
einsum('nqhd,nkhd->nhqk')	Scores: (N, heads, query_len, key_len).
Source mask	(N,1,1,source_len); broadcasts over heads and query positions.
Target mask	Lower triangle, expanded to (N,1,target_len,target_len).
Output vocabulary projection	(N,target_len,vocab_size): logits, not token IDs.

Concrete output: x has batch 2; trg has length 8; model receives trg[:,:−1], length 7. With target vocabulary 10, output shape is (2,7,10). The notebook uses learned positional embeddings and a post-normalization block. Its target-mask function supplies causality but does not combine a separate target-padding mask.

Close the notes and retrieve

What exactly does T5 predict? Why is decoder-only training parallelizable across positions while free generation still proceeds token by token?

6 Prompting and prompt sensitivity

Source: Lec_23.pdf, especially pp. 2-7 and 8-29.

Technique	Main idea	Distinguish it from
Few-shot / ICL	Place examples in the prompt to demonstrate a task.	Fine-tuning updates weights; ordinary ICL does not.
Chain-of-thought (CoT)	Use intermediate reasoning steps in demonstrations or output.	A long explanation alone does not prove correctness.
Self-consistency	Sample multiple reasoning paths and aggregate final answers.	A single greedy reasoning path.
Tree-of-thought (ToT)	Branch, score candidate paths, and backtrack.	One linear path; often requires more computation.
Graph-of-thought (GoT)	Combine and refine thoughts with graph connections.	A tree cannot merge separate branches into one node.

Sensitivity is different from accuracy

Intent-preserving variations ask the same thing with different wording, formatting, templates, or demonstrations. A model can score well on one benchmark prompt yet change behavior markedly when that prompt is rephrased. Correctness and sensitivity are separate properties.

Four aspects emphasized in the lecture

Response diversity: how varied the responses are.
Response distribution entropy: how broadly probability is spread across responses.
Semantic coherence: whether outputs preserve similar meaning.
Confidence variance: whether likelihoods change across equivalent prompts.

POSIX means PrOmpt Sensitivity IndeX. For N intent-aligned prompts x_i and their responses y_j, compare each response's likelihood under its original prompt x_j and alternative prompts x_i. Average absolute log-likelihood ratios, normalized by response length L_j; then average across prompt sets in the dataset.

Sensitivity of one set = $[1 / N(N-1)] \times$ sum over $i \neq j$ of
$[\ |\log(P(y_j|x_i) / P(y_j|x_j))| / L_j\ ]$
POSIX = mean set sensitivity across the dataset; higher means more sensitive.

Lecture findings to recognize

The reported experiments show that larger models are not automatically less sensitive; alignment can reduce sensitivity. Prompt-template changes are especially influential for the studied MCQ setting, while paraphrasing is especially influential for the studied open-ended setting. These are experimental findings from the slides, not universal guarantees for every model.

Close the notes and retrieve

Can two models have similar accuracy and different sensitivity? How does self-consistency differ from beam search? Which operation makes GoT more general than ToT?

7 Alignment and reward models

Source: Lec_24.pdf, especially pp. 2-5, 13-29.

Training stages: pretraining learns language patterns; instruction tuning learns to respond to task instructions; preference alignment encourages outputs people prefer. Instruction-following alone does not reliably resolve what should or should not be said.

Map text generation to reinforcement learning

RL term	LLM interpretation
State s_t	Prompt plus tokens already generated.
Action a_t	Generate the next token.
Policy $\pi_\theta(a_t \mid s_t)$	Probability distribution over next tokens.
Reward $r(x,y)$	Human feedback or a learned score for a response.
Next state	Append the generated token to the current context.

How a reward model learns

Collect triples (prompt x, preferred answer y+, rejected answer y−). Train a scalar scoring function r(x,y). A reward score is not necessarily a probability; convert score differences to pairwise preference probabilities.

$$P(y+ \text{ preferred to } y- \mid x) = \text{sigmoid}[r(x,y+) - r(x,y-)]$$
$$\text{Reward-model loss} = -\log \text{sigmoid}[r(x,y+) - r(x,y-)]$$

Worked example: preferred score 3, rejected score 1. Difference=2; sigmoid(2)≈.881. If scores are equal, preference probability is .5. Swapping the two scores yields about .119.

Why the reference model matters

$$\text{Maximize } J(\theta) = E_{y\sim\pi\theta}\,[r(x,y)] - \beta\, KL(\pi\theta(\cdot|x) \,||\, \pi\text{ref}(\cdot|x))$$

The reward term favors desirable responses; KL discourages excessive movement from a fixed reference policy. This helps limit reward hacking and loss of useful behaviors. Increasing β strengthens the penalty for the same divergence; β=0 removes that reference penalty.

RLHF versus RLAIF: the key distinction here is whether the preference judgments come from humans or AI. Constitutional AI uses written principles to guide AI critique/judgment. A reward model approximates preferences and can be wrong.

DPO scope: the uploaded Part-I taxonomy names it as an offline contrastive method; Parts I-II chiefly develop reward maximization and PPO. Recognize that DPO directly uses preference pairs without an explicit separately trained reward model in its standard formulation; spend derivation time on the supplied PPO material first.

Close the notes and retrieve

Why does instruction tuning not finish alignment? What is the reward model trained on? What happens when β increases?

8 REINFORCE, value, advantage and PPO

Source: Lec_25.pdf, especially pp. 5-27.

The log-derivative trick

$$\nabla_\theta \pi_\theta(y|x) = \pi_\theta(y|x)\, \nabla_\theta \log \pi_\theta(y|x)$$

For a fixed reward: $\nabla_\theta \mathbb{E}[r(y)] = \mathbb{E}[r(y)\, \nabla_\theta \log \pi_\theta(y|x)]$

This turns a sum over an enormous output space into an expectation that can be estimated from sampled responses. Sequence log-probability is the sum of token log-probabilities. A single sampled response can give a noisy learning signal.

Quantity	Meaning	Numerical example
$V(s)$	Expected return from a state under the policy.	$V(s)=3$
$Q(s,a)$	Expected return after taking an action, then following the policy.	$Q(s,a)=5$
$A(s,a)=Q(s,a)-V(s)$	How much better this action is than the state's average.	$A=5-3=2$

A state-dependent baseline can reduce variance without changing the expected policy gradient. In deterministic text transitions, $Q(s,a)=r(s,a)+\gamma V(\text{next state})$, so a one-step advantage estimate is $r+\gamma V(\text{next})-V(\text{current})$. A learned value estimate is approximate, not a guarantee of perfect credit assignment.

Why use an old policy?

Generating new responses after every update is expensive. Sample a batch from an older policy, then reuse it for updates. Importance ratios account for changed action probabilities.

$$\rho_t = \pi_\theta(a_t|s_t) / \pi_{\text{old}}(a_t|s_t)$$

$$L_{\text{clip}} = \mathbb{E}[\min(\rho_t A_t, \text{clip}(\rho_t, 1-\varepsilon, 1+\varepsilon) A_t)]$$

PPO maximizes this clipped surrogate. It limits the incentive for large changes; clipping is not a hard constraint guaranteeing every ratio stays inside the interval. The old rollout policy and the fixed reference model have different jobs.

Two sign-sensitive calculations

Let $\varepsilon=.2$. With $\rho=1.5$ and $A=+2$: unclipped=3; clipped=2.4; objective contribution=2.4. With $\rho=.5$ and $A=-2$: unclipped=-1; clipped=-1.6; minimum=-1.6. Always multiply before choosing the minimum.

Pipeline: sample prompts/responses → score rewards and KL → estimate returns/values/advantages → update policy with clipped objective → fit the value function → refresh rollouts. The KL reference penalty and PPO clipping are separate stabilizing mechanisms.

Close the notes and retrieve

Define V, Q and A. Why is REINFORCE noisy? What is in the PPO ratio denominator? What does clipping actually limit?

9 KG completion and evaluation

Source: Lec_27.pdf, pp. 2-13.

Knowledge graph: entities linked by relations, represented by triples (subject, relation, object). Completion predicts missing facts, for example (Paris, capital_of, ?). Embeddings turn entities/relations into vectors or other geometric objects; a scoring function ranks possible facts.

Training and score polarity

High-is-better score: P(o|s,r) = exp f(s,r,o) / sum over o′ of exp f(s,r,o′)
Margin loss = max(0, margin + f(negative) − f(positive))

Negative sampling corrupts the subject or object of a positive triple. It reduces computation but may label a real, missing fact as negative because KGs are incomplete. Missing does not necessarily mean false. For distance scores where lower is better, the margin-loss sign reverses.

Ranking metrics

MR = mean(rank_i), lower is better.
MRR = mean(1/rank_i), higher is better.
Hits@K = fraction of target answers ranked in the top K, higher is better.

Worked calculation: target ranks [1,2,4] → MR=7/3≈2.333; MRR=(1+.5+.25)/3≈.5833; Hits@2=2/3≈.6667. Do not take 1/MR: it is not MRR.

Raw versus filtered evaluation

A query can have multiple valid answers. In standard filtered link-prediction evaluation, remove other known true answers from candidates while keeping the particular target under evaluation. This avoids penalizing a model for ranking another true fact ahead of the target.

Example: target rank=5, but two candidates ahead are other known true answers. After filtering those candidates, target rank=3, so reciprocal rank improves from .2 to 1/3. Never filter out the target itself.

Additional lecture details

There are $|E|^2|R|$ possible triples if subjects and objects both range over E entities. The full softmax over entities is expensive, motivating sampled approximations. Dataset names to recognize: WN18, FB15k and their revised variants WN18RR and FB15k-237. For multiple gold answers, MAP averages precision at relevant ranks; do not confuse it with reciprocal rank.

Close the notes and retrieve

What does negative sampling assume? When can it produce a false negative? Explain filtered evaluation using a rank example.

10 Translation and rotation models

Source: Lec_28.pdf pp. 3-10.

Model	Geometry	Main issue / strength
TransE	s + r ≈ o in real space.	Simple and fast; struggles with multiple distinct objects for the same subject/relation.
TransH	Project entities to a relation-specific hyperplane, then translate.	Entities can have different effective representations for different relations.
RotatE	o ≈ s ⊙ r in complex space, with |r_d|=1.	Relations are rotations; handles important relation patterns through angle algebra.

TransE distance d = ||s+r−o||: smaller is better.
For distance scores: loss = max(0, margin + d_positive − d_negative).
RotatE distance = ||s ⊙ r−o||, with unit-modulus relation components.

TransE: calculate and explain the failure

s=(1,2), r=(2,−1), o=(3,1) gives distance 0. With o′=(5,1), L2 distance=2. Positive distance=0, negative=2, margin=1 gives hinge loss max(0,1+0−2)=0.

If Obama+attended must be close to both college A and college B, TransE pushes A and B close together. This explains the one-to-many limitation. For symmetric links s+r≈o and o+r≈s, an exact fit forces 2r≈0, limiting nontrivial symmetric modeling.

RotatE: learn the relation patterns

Pattern	Constraint / interpretation
Symmetry	r ⊙ r = 1. Each phase can be 0 or π modulo 2π; a nontrivial example is 180°.
Inverse relations	r_inverse = conjugate(r), since |r|=1. Reverse the angle.
Composition	r3 = r1 ⊙ r2: phases add.
Asymmetry	Use phases that do not make a relation its own inverse.

Worked complex example: entity s=1+0i; relation r=i is a 90° anticlockwise rotation; object s×r=i. Its inverse relation is −i. Applying i twice gives −1 (180°); applying it four times returns 1.

Trap: confidence scores and distances have opposite orderings. State whether high or low is better before choosing a loss or ranking. The slides use distance-style TransE/RotatE scores.

Close the notes and retrieve

Why does TransE struggle with one-to-many? How does a rotation represent inverse and composed relations?

11 Temporal KGs and hierarchies

Sources: 04_Knowledge_6_Temporal.pptx.pdf; Lec_34.pdf.

Facts can change with time

A temporal fact can be (subject, relation, object, start, end). Link prediction fills a missing entity with time known. Time prediction fills the timestamp or interval with the entities/relation known. A fact can be correct in one period and wrong in another.

Temporal idea	What to remember
HyTE	Time is represented by hyperplanes; use time-dependent projections.
TimePlex	Combine embedding plausibility with relation-pair time-gap and recurrence signals.
Relation gaps	A birth date should be consistent with graduation or marriage dates; model gap statistics.
Recurrence	Repeated events provide signals through the gaps to nearby occurrences.
Interval evaluation	Overlap alone can miss near-versus-far errors when intervals are disjoint; the slides discuss interval-aware measures.

Example: Obama’s US presidency interval 2009-2017 can support a temporal link query. A proposed birth date after an already-known marriage date is implausible. Time-gap constraints provide information absent from isolated triple scoring.

Hierarchies require containment or order

A hierarchy can be a directed acyclic graph. “camel is-a mammal” and “mammal is-a animal” imply “camel is-a animal”: transitivity. General graph proximity alone does not fully express ancestor-descendant direction.

Representation	Core idea
Hyperbolic / Poincaré	Fit tree-like growth into negatively curved geometry. Generic concepts tend toward the center, specific concepts toward the boundary.
Order embeddings	In the lecture convention, x descendant-of y implies $u_x \geq u_y$ elementwise. Check the convention before applying signs.
Box embeddings	Represent concepts as boxes; a child box is contained in the parent box. Intersections express overlap.
Tree traversal intervals	Nested entry/exit intervals encode ancestry: I(child) is contained in I(parent).

Checks: interval [2,3] is inside [1,10], so it can represent a descendant. For u_child=(3,4) and u_parent=(2,3), the elementwise order holds. Distortion measures maximum distance stretch divided by minimum stretch; lower distortion better preserves relative graph distances. Near the Poincaré boundary, hyperbolic distances can grow very large.

Close the notes and retrieve

Distinguish temporal link prediction from time prediction. Why can hierarchy modeling need something beyond ordinary Euclidean proximity?

12 Compression and efficient adaptation

Source: Lec_30.pdf, especially pp. 11-12, 22-40, 42-59.

Method	What changes?	Exam distinction
Quantization	Reduce bits used for numerical values.	Same nominal parameter count can need fewer storage bytes.
Pruning	Remove or zero selected weights/components.	Fewer nonzero weights does not guarantee faster execution.
Distillation	Train a student using a teacher’s predictions.	Knowledge transfer into another, usually smaller, model.

Quantization

PTQ: quantize after training; static activation quantization often uses calibration examples. QAT: model training simulates quantization effects. LLM.int8(): isolate important outlier computations into higher precision while quantizing the rest.

Simple affine convention: q = clip(round(x/scale)+zero_point)
Reconstruction: x_hat = scale × (q−zero_point)
Raw weight storage = parameter_count × bits_per_parameter / 8 bytes

Memory example: 1 billion weights at FP32 ≈4 GB; FP16≈2 GB; INT8≈1 GB; 4-bit≈0.5 GB in decimal units. These are weight-only estimates: add scales, other metadata, activations, KV cache and training states as applicable. FP32→INT8 is about a 4× raw weight reduction, not a promise of 4× speed.

QLoRA: remember all four ingredients

NF4: a 4-bit format designed for normally distributed weights. Double quantization: quantize quantization constants. Paged optimizers: manage memory spikes. LoRA: train low-rank adapters on a frozen quantized base. QLoRA does not mean training every base weight at 4 bits.

Pruning

Magnitude: remove small absolute weights. Wanda: use weight magnitude together with input activation information. Unstructured: individual weights, often requiring sparse kernels for speed. Structured: heads, channels or blocks; match hardware-supported structures. A 2:4 sparsity pattern keeps two values out of each supported group of four.

Distillation

Soft targets preserve the teacher’s probability distribution; hard targets retain only one chosen class/token. Token-level distillation transfers distributions at token positions. Sequence-level distillation uses teacher-generated complete output sequences. A teacher can be wrong; distillation does not guarantee superior accuracy.

Close the notes and retrieve

Quantization versus pruning versus distillation? What does double quantization quantize? Why might sparse weights not reduce latency?

13 Residual stream and interpretability

Sources: Lec_31.pdf; Lec_32.pdf, especially pp. 3-11, 14-31, 50-61.

Residual stream: a shared representation updated by attention and feed-forward outputs through vector additions. The final representation is mapped to vocabulary logits by an unembedding matrix. This perspective helps attribute contributions and trace information flow.

QK circuit: $W_{QK} = W_Q W_K^T$ → influences where a head attends.
OV circuit: $W_{OV} = W_V W_O$ → transforms information written back.
Residual update (schematic): x_next = x + attention_update + FFN_update.

The schematic suppresses normalization and the fact that FFN usually reads a state already updated by attention. With row-vector conventions, the final unembedding has shape d_model × |V|. Successive heads can compose: later heads read information written by earlier heads.

Technique	Question / operation	Main caveat
Probe	Train a classifier on fixed representations to decode a property.	Decodable information is not proof that the model causally uses it.
Control task / selectivity	Compare linguistic-task performance against a control-label task.	A powerful probe may memorize labels or learn too much on its own.
Activation patching	Replace chosen activations between clean/corrupted runs and measure output change.	Results depend on corruption, intervention and metric.
Logit lens	Project intermediate residual states through the unembedding into vocabulary space.	Intermediate representations may not align with final decoding space.
Sparse autoencoder	Reconstruct activations from sparse combinations of learned dictionary features.	Features and labels are approximations; individual neurons can mix concepts.

Activation patching: know the three runs

1. Clean: run “The Eiffel Tower is in” and cache activations; expected answer Paris. 2. Corrupted: use a changed input and record output. 3. Patched: inject a selected clean activation into the corrupted run. If the Paris score recovers, that component has evidence of a causal contribution under this test.

Mechanistic interpretability: reverse engineer the computations implementing behavior. A circuit is a relevant computational subgraph. Superposition: many features may share neural dimensions; sparse dictionary learning tries to separate useful directions. Attribution patching approximates intervention effects with gradients for efficiency; activation patching runs actual replacements.

Close the notes and retrieve

QK reads where; OV writes what. Explain why a successful probe is weaker causal evidence than a well-designed patching intervention.

14 Responsible LLMs

Source: Lec_36.pdf, especially pp. 4-6, 8-9, 15-16, 19-25, 27-48.

Four dimensions in these slides

Dimension	Meaning
Explainability	Make model behavior and internal workings understandable.
Fairness	Identify and mitigate harmful biases across people and groups.
Robustness	Remain reliable under unusual or perturbed inputs.
Safety and security	Resist malicious use and attacks; reduce harmful behavior.

Bias: sources and measurement

Bias can reproduce social stereotypes through training-data selection, outdated language usage, and overrepresentation of high-resource languages and cultures. It can encourage discrimination and reduce trust. Good language modeling scores do not by themselves establish fairness.

StereoSet domains: gender, profession, race, religion. LMS: preference for meaningful over meaningless associations. SS: preference for stereotypical over anti-stereotypical associations; ideal balance is 50, not 0 or 100.

Using percentage units: ICAT = LMS × min(SS,100−SS) / 50

Worked example: LMS=80, SS=75 → ICAT=80×25/50=40. With the same LMS and SS=50, ICAT=80. A low SS can indicate reverse stereotypical preference; the metric rewards balance.

Regard is not ordinary sentiment

Sentiment describes positive/negative tone. Regard captures the social perception of a described demographic group. A sentence can sound positive about a person sustaining a livelihood yet describe a low-status stereotype; the two labels need not agree.

Two mitigation approaches covered

Adversarial triggers: optimize a token prefix to shift associations, for example toward positive/neutral and away from negative regard for a group. A trigger can also amplify bias; its effect depends on the objective.
In-context safety demonstrations: retrieve relevant examples and include them in the prompt. The slides compare random, BM25 and SentenceTransformer retrieval, then condition generation on selected demonstrations.

Practical distinction: ordinary prompting/ICL changes the context; fine-tuning changes model parameters. Alignment can improve measured behavior but does not prove bias has disappeared or that all prompts are safe.

Close the notes and retrieve

Name all four dimensions. Explain temporal bias versus temporal KG modeling. Calculate ICAT and explain why SS=50 is the balanced target.

15 Closed-book practice

Use this after the first pass. 30-40 minutes. Answers are on the next page.

Give each question 1 point. This is a revision diagnostic, not a prediction of the official exam. Write a number, formula or one-sentence explanation before looking at answers.

01. For corpus cat×10, bat×5, bag×12, tag×4, cats×5, what is the initial frequency of pair (a,t)?

02. WordPiece scores: pair A has counts (pair=12, a=20, b=30); pair B has (pair=6, a=5, b=10). Which merges first?

03. For probabilities [.5,.25,.15,.1], which candidates survive top-p=.8, and what are the new probabilities?

04. Name the difference between beam size k and top-k sampling.

05. What happens to a distribution as positive temperature increases? Does positive temperature change greedy argmax?

06. Write scaled dot-product attention, including a causal mask.

07. For 512 model dimensions and 8 heads, what is each head’s dimension?

08. Why is masking a forbidden score with 0 insufficient?

09. For Q shape (2,8,7,64) and K shape (2,8,9,64), what is QK^T shape?

10. What is every sinusoidal sine/cosine pair at position zero?

11. What does RoPE do to Q and K, and why do relative positions enter their dot product?

12. Name the distinguishing pretraining targets of BART and T5.

13. In the uploaded notebook, batch=2, original target length=8, vocabulary=10, input trg[:,:−1]. What is output shape?

14. Explain why prompt accuracy and prompt sensitivity are different.

15. What operation lets GoT represent something a tree cannot?

16. Reward scores 2 and 2: what is Bradley-Terry preference probability?

17. V(s)=4, Q(s,a)=7. Compute the advantage and interpret its sign.

18. PPO: ε=.2, ratio=1.4, A=+3. Compute the clipped surrogate contribution.

19. Which policy is used in the denominator of the PPO ratio? What is the separate role of the reference model?

20. Ranks [1,2,4]: compute MRR and Hits@2.

21. A target ranks 6th; three preceding candidates are other known true answers. What is its filtered rank?

22. TransE distances: positive=1, negative=1.5, margin=1. Compute the hinge loss.

23. In RotatE, what is the inverse of a 90° relation? What is two 90° relations composed?

24. Distinguish temporal link prediction and time prediction; name two TimePlex temporal signals.

25. Explain why a hierarchy can be modeled using nested boxes. Under the lecture order convention, is (4,5) a valid descendant of (3,3)?

26. Estimate weight-only storage for 2 billion weights at FP16 and INT8 in decimal GB.

27. Name QLoRA’s four ingredients. Distinguish structured and unstructured pruning.

28. Which circuit controls attention matching and which transforms information written to the residual stream?

29. Why does successful probing not establish causal use? Describe a patching test.

30. LMS=90, SS=60. Compute ICAT, and name the four dimensions of responsible LLMs.

16 Answer key and final review

01. 20: cat, bat and cats contribute 10+5+5.

02. B: .12 versus A’s .02. Highest ratio wins in the lecture formulation.

03. First three; .556, .278, .167 (renormalize by .9).

04. Beam retains partial sequences by cumulative score; top-k sampling samples the next token from k candidate tokens.

05. Flatter distribution; no argmax change for positive T, except tie-handling details.

06. softmax(QK^T/√d_k + M)V, with future-token M entries −∞.

07. 64.

08. exp(0)=1, so it can still receive softmax probability.

09. (2,8,7,9).

10. (0,1).

11. Position-dependent rotations; common shifts preserve relative angle/dot product.

12. BART reconstructs full clean text; T5 span corruption predicts removed spans with sentinels.

13. (2,7,10), output logits.

14. Correctness on selected prompts versus stability across equivalent prompts.

15. Merging/aggregating branches, as well as iterative refinement.

16. .5.

17. +3; better expected return than the state’s policy average.

18. min(4.2,3.6)=3.6.

19. Old rollout policy; the reference model anchors KL regularization.

20. MRR=7/12≈.5833; Hits@2=2/3.

21. 3.

22. max(0,1+1−1.5)=.5.

23. −90° (270°); composition gives 180°.

24. Known time/missing entity versus known triple/missing time; relation-pair gaps and recurrence.

25. Containment represents subtype/ancestor relations; yes, (4,5)≥(3,3) elementwise.

26. 4 GB and 2 GB respectively, excluding overhead.

27. NF4, double quantization, paged optimizers, LoRA. Structured removes supported groups/components; unstructured selects individual weights.

28. QK controls matching; OV transforms information written.

29. A probe shows decodability; intervention is needed to test use. Replace a corrupted activation with a clean one and measure restoration.

30. 90×40/50=72. Explainability, fairness, robustness, safety/security.

Use the score to choose your next block

24-30: check explanations and redo wrong calculations. 18-23: revisit weak pages and repeat those questions without the key. Below 18: spend the next block on tokenization, decoding, attention and training families before retrying. These are study thresholds chosen for this pack, not official passing cutoffs.

Final 25-minute recall sheet

0-5 minutes: foundations

BPE = frequent pair. WordPiece = lecture frequency-ratio merge, longest-match segmentation. Unigram = probabilistic segmentation and vocabulary pruning. Beam = sequence search; top-k/top-p = token sampling. Temperature acts on logits.

P(sequence) = product of conditional token probabilities.
Attention = softmax($QK^T/\sqrt{d_k} + M$)V.
Causal: allow j≤i; future logits receive −∞.
head_dim = d_model / heads.

5-10 minutes: architecture and training

Self-attention QKV: same sequence. Cross-attention: decoder Q, encoder KV. Residuals add updates. LayerNorm normalizes hidden features. Sinusoidal PE uses sin/cos; RoPE rotates Q/K. Encoder: bidirectional representations. Decoder: causal next-token generation. BART: full reconstruction. T5: missing spans. ICL: examples in context; SFT: update weights.

10-15 minutes: alignment

Preference probability = sigmoid(r_preferred−r_rejected).
Policy objective = expected reward − β KL(policy || reference).
Advantage = Q−V.
PPO ratio = new action probability / old action probability.
PPO = min(ρA, clip(ρ,1−ε,1+ε)A).

Human preferences → reward model → policy optimization. Q/V estimate returns; advantage compares to baseline. KL and clipping have separate purposes. Clipping limits optimization incentive, not every actual ratio.

15-20 minutes: knowledge and efficiency

MRR = average reciprocal ranks; Hits@K = fraction with rank≤K.
TransE: s+r≈o. RotatE: s⊙r≈o, |r_d|=1.
Weight bytes = parameters × bits / 8.

Filtered ranks remove other known truths, keep target. Temporal facts add time; TimePlex uses gaps and recurrence. Hierarchies need order/containment; Poincaré geometry suits trees. Quantize bits, prune parts, distill teacher behavior. QLoRA: NF4 + double quantization + paged optimizers + LoRA.

20-25 minutes: interpretation and responsibility

QK = matching; OV = writing. Probe = decodable; patching = intervention. Logit lens decodes intermediate states; SAE separates sparse features. Responsible dimensions: explainability, fairness, robustness, safety/security. SS balance=50. ICAT=LMS×min(SS,100−SS)/50. Regard differs from sentiment. Safety demonstrations can be retrieved into context.

In the exam: check whether a score is a distance or confidence, whether probabilities need renormalization, whether a question allows multiple answers, and whether the requested metric is a mean rank or mean reciprocal rank. Follow the actual paper's marking and time instructions.

Source note: definitions and topic scope are synthesized from the supplied lectures. Worked practice examples, study allocations and diagnostic thresholds are author-created. Historical model sizes, prices and speculative architecture claims were not used to predict current systems or exam weightings.